

DHATUKI.

DEPARTMENT NEWSLETTER 2023-24

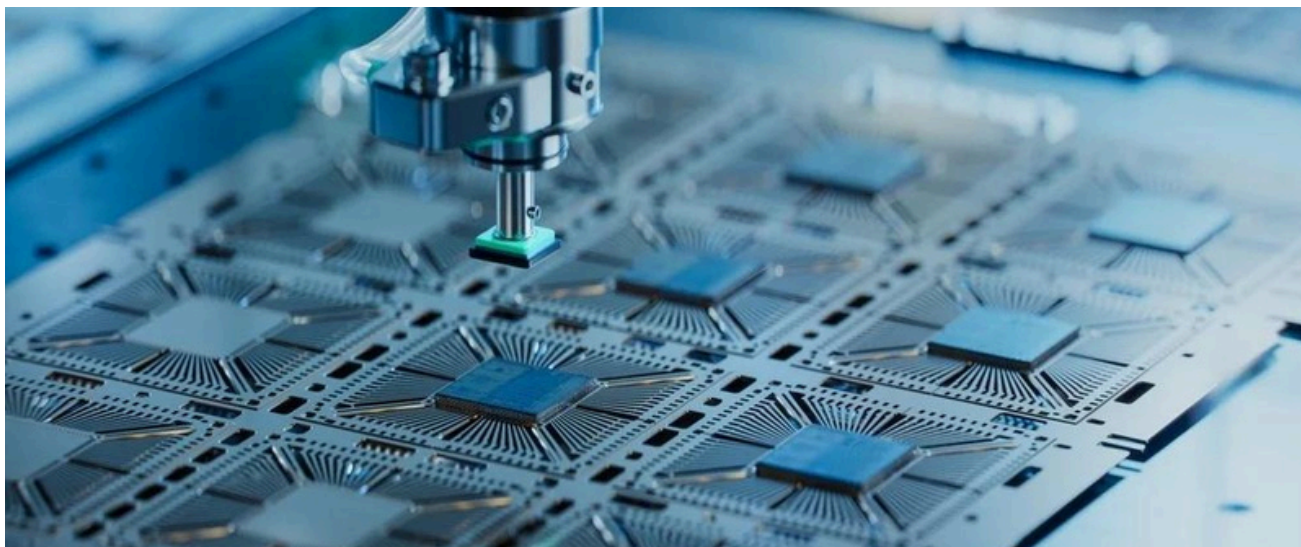
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IIT BOMBAY'S

**DEPARTMENT OF
METALLURGICAL
ENGINEERING &
MATERIALS SCIENCE**



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HOD's NOTE

Dr. Viswanathan N Nurni

We are excited to announce the latest release of Dhatuki, the annual newsletter of the Department of Metallurgical Engineering and Materials. This edition is a testament to the hard work of our team. As we reflect on centuries of technological progress, it is evident that the development of materials science has been a driving force. Looking ahead, material innovation will continue to shape future inventions. However, this fact is not always apparent, especially to those new to the field. This is why I believe the Dhatuki team is doing an exceptional job of highlighting the importance of materials through their thought-provoking writings. We would like to extend a big shoutout to the team for their outstanding work.

Research in Electronic Materials and Semiconductors @MEMS

By: Darshan Mali

Fourth-year Undergraduate

Since time immemorial, advancements in materials science have been the driving force for human progress. Materials science stands as a pillar of scientific innovation, enabling advancements across various industries. Among the myriad of fields that benefit from materials science, semiconductor technologies stand at the forefront of technological innovation, driving advancements that touch every aspect of our modern lives. Semiconductors, the foundation of modern electronics, have revolutionized the world by powering our smartphones, computers, and countless other devices.

At the heart of electronic devices are materials carefully engineered for their specific properties. The relevance of ongoing research in this field extends beyond the confines of traditional electronics, influencing diverse sectors such as healthcare, renewable energy, and computing. The potential of semiconductor research is vast, offering opportunities for groundbreaking discoveries, technological revolutions, and solutions to pressing global challenges.

The potential inherent in semiconductor research lies in its ability to push the boundaries of what is technologically possible. Advancements in materials science, such as the exploration of novel materials like graphene and quantum dots, open avenues for developing faster, smaller, and more energy-efficient devices. Quantum computing, a field within semiconductor research, has the potential to revolutionize computation, solving complex problems that are currently beyond the reach of classical computers.

The dynamic nature of semiconductor research presents numerous opportunities for breakthrough discoveries. Researchers are continually exploring new materials, fabrication techniques, and device architectures. Current research is focused on developing new materials or enhancing existing ones to meet the ever-increasing demands of technology. Innovations range from the exploration of two-dimensional materials like graphene and transition metal dichalcogenides to the synthesis of complex organic compounds with unique electronic characteristics.

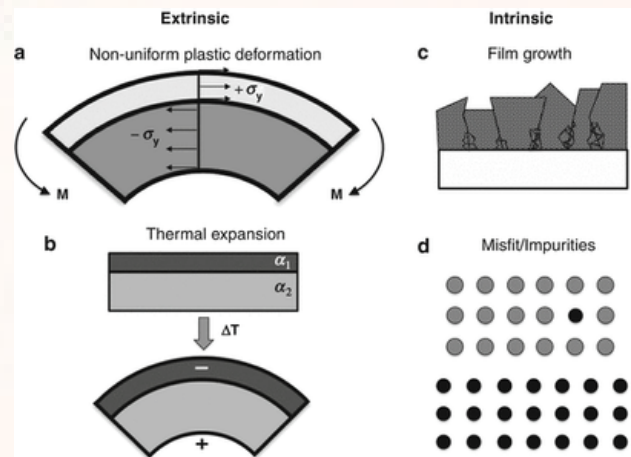
Like the rest of the world, researchers at our MEMS department are greatly enthused by this field and have been carrying out exciting research in our department labs. Given that electronic materials science is an extremely vast field, in this article, we try to explore the research done by one of the research groups in our department in detail. This article addresses some of the common questions that budding UG researchers have regarding this field with the help of one of our department professors working on electronic materials: **Prof. Rajiv O. Dusane** (Head of Semiconductors, Thin Films and Plasma Processing Laboratory), who shares his experience of doing research in semiconductors and thin films, and provides guidance for students to progress in this field.

MECHANICAL STRESSES IN THIN FILMS

BY SATYAM RATH
THIRD-YEAR UNDERGRADUATE

Thin film systems are used in many areas, such as in optical, electronic and mechanical applications. Emerging areas like nanotechnology are also based on thin film technology. Due to the development in vacuum technology and electrical power facilities, thin films are used in various industries to provide specific physical and chemical properties to the surface of bulk materials. Consequently, precisely analyzing such systems in the applications mentioned becomes essential. Fabrication techniques are designed to fabricate these thin films for a specific application. For instance, Molecular Beam Epitaxy (MBE) is used to fabricate thin films with precise thickness and composition control while offering in-situ measurements of various parameters like lattice constant, which enables us to draw conclusions relating to growth morphology. The term 'epitaxial' is applied to a film grown on top of the crystalline substrate in ordered fashion that atomic arrangement of the film accepts crystallographic structure of the substrate.

I applied for the Summer Undergraduate Research Programme (SURP) in July 2022 under the supervision of Prof. Tanushree Choudhary.



The initial problem statement was "Surfactants for Layered Semiconductor Synthesis." As is customary in any research endeavour, my project started with going through literature relevant to the project. Literature Review entails reading and understanding any literature relevant to the problem statement, allowing one to draw insights into the existing work done in that area, gain knowledge essential for the research, and develop an area to work on based on specific questions which have not been addressed effectively.

Additionally, an important aspect of literature review involves documentation which may be useful later on. After reading several papers on surfactant-mediated epitaxial growth and understanding the thin film growth process, I learned the importance of mechanical stresses in thin films and their importance in altering various properties.

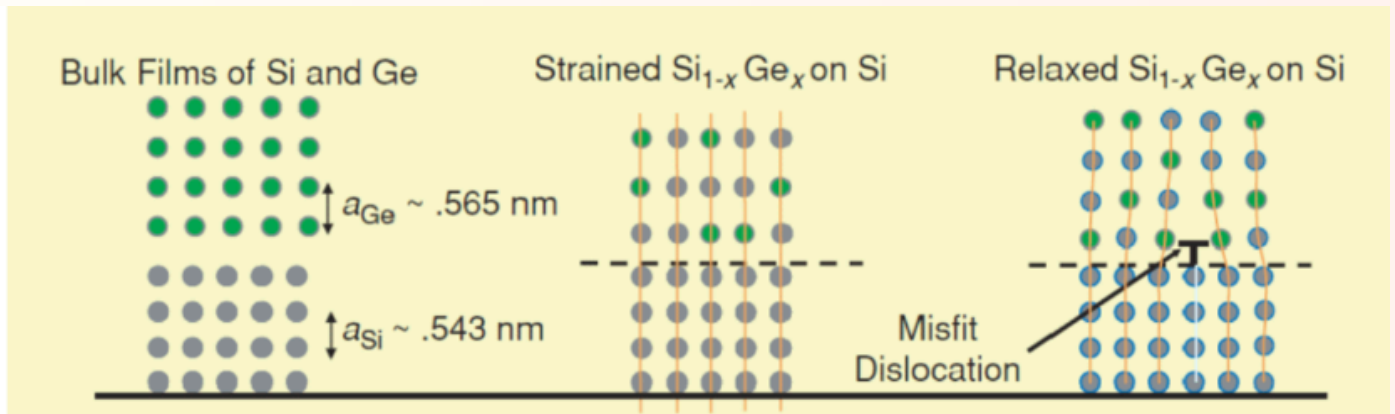


Figure 1. Depicts strained $Si_{1-x}Ge_x$ on Si due to lattice mismatch and misfit dislocations relieving the stress.

Strains in hetero-epitaxial systems alters its electronic and optoelectronic properties because of the difference in the lattice parameters of the film and the substrate. Thus, they can be employed in the exciting field of bandgap engineering. However, in certain applications, the existence of residual stresses can be detrimental, causing the films to delaminate or crack. Thus, a sound understanding on how these stresses develop becomes essential. Consequently, I started working on Mechanical Stresses in Thin Films. As described by Pramanik (2011), the residual stresses in thin films are a consequence of the coupling of thermal mismatch, lattice mismatch and misfit dislocations at the interface. At the beginning of the film growth, elastic energy, stored in the film due to lattice mismatch, is too small for nucleation of dislocations. As the film thickness is growing up dislocations begin to nucleate and it partially relaxes the strain accumulated in the film. Using COMSOL, I simulated the stresses in thin film systems as a consequence of the difference in the Coefficient of Thermal Contraction of the film and the substrate.

This exercise aimed to perform FEM simulations and compare it to experimental findings. The thin film system is cooled from the deposition temperature to room temperature by convection, and they contract at different rates resulting in residual stresses. The von mises stress extracted from several positions in the film with respect to the interface revealed the independence of residual stresses with respect to distance from the interface.

Currently, I am working on realizing a lab setup for fabricating films on flexible substrates, which is prominently used in flexible electronics and subjecting it to various forms of mechanical deformation, while studying the alterations in its properties.

At the beginning of my research, everything felt intimidating. The idea of reading multiple papers, tackling concepts that I had never heard about felt challenging. Eventually, that initial friction went away and I could make progress. Attending meetings, and giving presentations on my research was an incredible learning experience which has enabled me to convey my ideas in an effective fashion.



SEMICONDUCTOR THIN FILMS AND PLASMA PROCESSING LAB

PROF. RAJIV O. DUSANE

Can you give a brief overview of the research done in your lab?

Over the last 30 years I have focused on the low thermal budget synthesis of Silicon based thin films for application in photo-voltaics, detectors, light emitting diodes and other semiconductor thin film devices. Our main contribution is the establishing of the Hot Wire Chemical Vapor Deposition (HWCVD) for this synthesis. In the last 10 years we have ventured into the synthesis and application of the Silicon Nanowires (again synthesized by the HWCVD) with great success in the energy storage landscape.

What was your motivation to pursue research and what do you look for in students that want to pursue research?

In my early college days itself I was fascinated by various technological developments and had almost decided to pursue research, though the exact area was not clear. During those days the conversion of sunlight into electricity ie solar cells was being promoted since people realized that the fossil fuels were not a permanent source of energy and this photovoltaic conversion is also ecologically highly beneficial.

I was also lucky to get an opportunity to join the group of the eminent scientist Prof. V. G. Bhide to do research in this area. Our group was one of the three groups in the country to initiate research in developing Amorphous Silicon thin film solar cells.

Pursuing research must be internally driven and deeply rooted in the mind of the student. Research is nothing but journey undertaken to unravel the mysteries of the nature and make them useful for the utilization for mankind. Applied research leading to technology development is a natural outcome of understanding fundamental concepts. So a researcher should have passion to explore such natural phenomena and convert them to applications. This calls for a huge commitment and endurance but there is lot of satisfaction and joy in the end.

Materials Science consists of a wide array of fields. How did you decide which field of Materials Science you wanted to pursue research in and how can students make that decision?

As I mentioned earlier I was only determined to do research. My exposure to solid state physics and then a project using Ics and Opamps led to an inclination towards semiconductors. And finally the opportunity also came my way which allowed me undertake research in semiconductor thin films.

In the early eighties we did not have the internet to learn about the various developments as we have now today. Today a student interested in research can make a decision by learning the latest developments in an area which interests him. One should also either have some background training in that area or should be ready to take fundamental courses in that area so that he appreciates the underlying principles which drive the innovations in that field.

The current era is quite often regarded as the “Silicon Age” or the “Digital Age”. What is the role of Materials Science in your research and the current era being referred to as the “Silicon Age” or the “Digital Age”?

Silicon is the back bone of the digital age and could remain so for a long time. Though this is true, one should appreciate that the digital age is a collective outcome of the semiconductor manufacturing and the logic circuit designing. Materials contribute to former. If you look deeper, you will realize that the semiconductor industry uses almost all classes of materials from sand, metals, polymers, ceramics and so on. Every aspect of these materials is important for its optimum utilization in the device. Semiconductors are the heart as they comprise the active part. The metal is like the arteries and veins of the device as they carry the current. The polymers are the support and protection as they form the packaging and so on. So you see the relevance of materials science in the semiconductor industry.

How do you envision the development of semiconductors shaping the future, both in terms of technological advancements and societal impact?

One of the key technology driving economical growth and societal development is the digital

technology which is supported by the semiconductor technology. The additions of Internet of things (IoT), Artificial Intelligence and Machine Learning (AI and ML) add more dimensions calling for interdisciplinary expertise among the workers and researchers. The scope is expanding exponentially and so the impact. Exploration of outer space calls for newer materials within this framework which could function efficiently in the harsh atmosphere of outer space. All these will increase the demand for knowledge and expertise in all aspects of materials from developing new materials to new processes.

What are the current career opportunities in the field?

As mentioned earlier the horizon that materials encompass is so large that there is no dearth of opportunities. However, the level of expectations has increased a lot as the technology is becoming very complex. Aspirants should focus on expertise and specialization as all other mundane jobs will be taken care by AI and robots. But keep in mind that developing better digital and robotic technologies is also materials intensive and so opportunities for good materials scientists are abundant.

Do you have any advice for students who are interested in getting into research?

Key aspects- Make your mind whether you are truly interested in exploration and be aware of the stringent requirements in terms of academic discipline and endurance.

One must have the passion to become a successful researcher. Develop the ability to observe and correlate the observations meaningfully. I would like to state that research in materials science and technology is not only very exciting but also very rewarding.

PEROVSKITES: A RISING STAR IN OPTOELECTRONIC APPLICATIONS

BY Koushika Saravanan
SECOND-YEAR UNDERGRADUATE

In the ever-evolving landscape of semiconductor technology, one material stands out as a beacon of innovation: perovskites. From being a mineral's crystal structure to their perfectly tuned form of highly efficient solar cells, perovskites have emerged as a subject of intense research and exploration.

At its core, perovskite is a type of crystal structure characterized by its cubic arrangement of atoms with the form ABX_3 . When X is a halide anion, it is called halide perovskite, which finds extensive interest in real-world applications. In the Optoelectronics and Devices lab of our department, researchers work on all crucial applications of halide perovskites - apart from solar cells, LEDs, and photodetectors, their work also includes light-responsive materials that can be used as Photodetectors (sensors), for a futuristic window that reduces sunlight transmitted as applied voltage increases (electrochromism and thermochromism), to identify counterfeited currency notes (luminescence) and to reduce the problem of highly aromatic organic textile dyes suffocating the water bodies (dye degradation). The diverse applications of halide perovskites, as explored in our department's lab, demonstrate their remarkable versatility. Particularly for solar cells, Organometallic halide perovskites (OMHPs), where the A-site is an organic cation, are particularly promising candidates for the next generation of solar cells.

By altering the chemical composition and internal architecture of the crystal, researchers can manipulate the bandgap, absorption spectrum, and charge transport properties of the material and this unique organic-inorganic (hybrid) combination offers that exceptional tunability, better stability, and enhanced optical properties.

With our country setting an ambitious target of achieving 450 GW of renewable energy capacity by 2030, the need for innovative and efficient solutions becomes paramount. Perovskite solar cells, particularly OMHPs, offer a promising avenue to address this challenge. Their remarkable efficiency improvements showcase their potential to significantly contribute to achieving these goals.

The efficiency of a $MAPbI_3$ and $MAPbBr_3$ (MA - Methyl Ammonium) based perovskite solar cell in 2009 was just 3.81% but now for a solar cell based on perovskite the efficiency is a remarkable 33.9%. This represents a phenomenal 30.09% improvement in just 15 years, far surpassing the modest 13.6% efficiency gain achieved by mainstream silicon-based solar cells over the past 48 years.

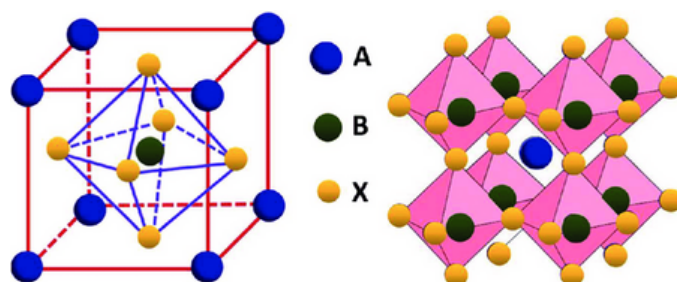


Figure 1: Perovskite Structure

Notably, the record-breaking 33.9% efficiency was achieved using a tandem solar cell configuration, where multiple absorber layers with varying bandgaps are stacked to capture a broader range of the solar spectrum, further enhancing energy conversion efficiency by reducing the thermalization and transmission losses endured during the process.

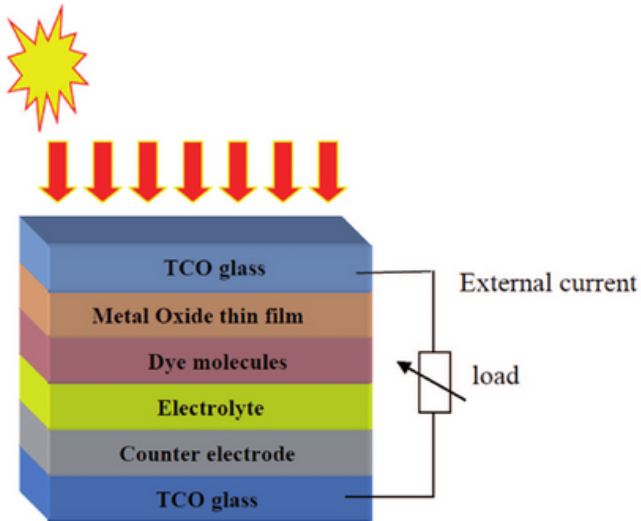


Figure 2. Structure of Dye-Sensitized Solar Cell

Balancing stability, efficiency, and production cost is crucial for entry into the solar cells industry. Perovskite solar cells offer lower production costs due to simpler fabrication, but they have high moisture and photo instability which is a challenge that the researchers are actively addressing today.

The steps for introducing OMHP solar cells into the industry have already begun. This month, the first report on the industrial roll-to-roll printing method for the production of hybrid perovskite solar cell modules marks the headway of this innovation from research labs into the industrial space and hopefully into the market soon.

Another promising field in which halide perovskites are making great strides is LED technology.

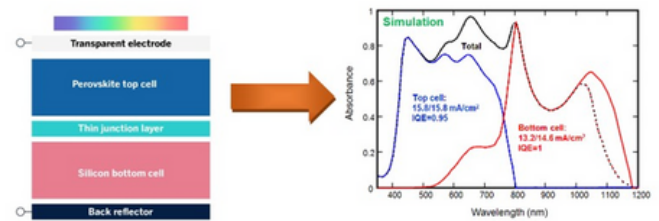


Figure 3: Example of improvement of range of wavelength absorbance with tandem cells (Graph is just for understanding)

Unlike conventional LEDs requiring high-temperature synthesis, perovskite LEDs can be fabricated at lower temperatures, reducing production costs. Their high efficiency, solution processability, tunable bandgap for versatile color emission, and easy deposition make them frontrunners for next-generation color displays, lighting solutions, and optical communication systems. Similar to solar cells, the primary hurdle for perovskite LEDs is limited operational stability. However, their low-cost and high-efficiency nature offers a compelling advantage.

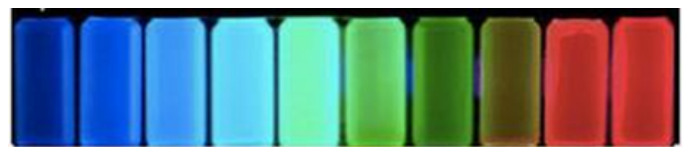


Figure 4: Organometallic Halide perovskites tuned to emit different colors

As we stand at the forefront of a renewable energy revolution, perovskite materials offer a beacon of hope and immense potential. However, their successful integration into practical applications necessitates overcoming challenges related to stability, efficiency, scalability, and achieving consistent manufacturing reproducibility. Despite these hurdles, the remarkable properties of halide perovskites continue to fuel the unwavering determination of researchers worldwide, propelling them toward a future brimming with innovative applications.



GRADUATE STUDIES AND CORE INDUSTRY OPPORTUNITIES IN MATERIALS SCIENCE

By: Mandar Gadre

The decision to pursue graduate studies is a pivotal moment in the lives of many students, and one that almost all undergraduates are rattled by at one point of time or another. It marks a significant crossroads where questions, doubts, and uncertainties abound. With numerous factors to consider, such as career prospects, financial implications, and personal aspirations, it's no wonder that students often find themselves confused when contemplating the path of higher education. We asked Mandar Gadre, an alumnus of our department, who pursued higher studies in Materials Science and Engineering to shed more light onto this topic, and to provide his insights to help students better navigate this complex terrain.

Graduate Studies in Core Tech: Start with the right WHYs

There are some great reasons to get into higher studies in core science and engineering branches:

Getting closer to the cutting edge:

A few decades ago, a Bachelor's degree was usually enough to take you close to the cutting edge of your field. Now with specialised research efforts in various areas and a rise in interdisciplinary studies, fields have progressed so considerably that everything cannot be covered in a 4-year university course. Graduate studies will be your chance to get closer to the cutting edge, to finding out where the limits of our current knowledge are, and enabling you to contribute in pushing them further.

Leverage the compounding effect:

Einstein famously said that the compounding effect is the most formidable force in the universe. Going deep into an area of interest, experimenting, collaborating, exploring unique angles, discovering new problems and then working on innovative solutions – all of this takes time. It also builds on prior experience. I recommend that you view it as a chain of knowledge links. The more you explore and work on important problems, the higher your chance of adding very valuable links. Through this, you stand a chance to make fundamental contributions, to stand on the shoulders of giants and also on your own prior work, experience, even mistakes.

Gold is in the depth:

Higher studies give you a chance to dig deeper and take on unsolved problems in your area of interest. There are loads of unsolved and interesting problems in every area: they present immense intellectual challenges and the opportunity to pursue a fulfilling career.

Know what real expertise looks like:

Graduate school in tech is one of the great ways to slowly but surely get to know what real expertise looks like in action. It is an opportunity to work with seasoned scientists, researchers, and professors honing their craft every day. If we are not exposed to researchers much better than ourselves, we risk thinking too highly of ourselves.

Fair warnings: Do NOT go for graduate studies for the wrong reasons:

- Don't just do it so you can get addressed as "Doctor", or to attach the coveted Ph.D. to your name.
- Don't just do it to go to a brand-name school – wherever it is.
- Don't just do it driven by Fear of Missing Out.
- Don't just do it because your close friends are doing it.
- Don't just do it to shine in someone's eyes, to fulfil someone else's expectations of you.
- Don't do it to postpone the hard decisions you need to make.
- Don't do it while you wait for something else to happen in your life.

These are hard-earned lessons – you won't find them in any book!

Graduate Education: What is a Ph.D.?

To understand what the higher education degrees – a Masters with thesis and especially a Ph.D. look like, let's use the framework by Matt Might:

"Imagine the whole of human knowledge in any given field as a series of concentric circles. At the very centre are the fundamentals and the long-known facts. As you move away from the centre, the rings represent the knowledge added by successive scientists-engineers-researchers. When you go through a Bachelor's programme, you gain a specialty – which is an extension of one of the inner rings. The outermost boundary of the circle represents the edge of the current knowledge – the 'state of the art'.

With a Master's degree, you deepen that specialty. A Ph.D. programme typically will bring you even closer to the cutting edge – and your thesis will add something noteworthy to the human knowledge on that subject. That's the little bump – that's your contribution, that's you pushing the envelope."

Opportunities after MS/ Ph.D.

Academia: Professorships (Research + Teaching), Serial Post-Docs

National Labs: Applied Research, for example in Dept. of Energy labs (USA), CSIR labs (India), etc.

Industry (Large): Engineering, Operations, Tech/R&D

Industry (Start-ups): Extreme Cutting Edge, Interdisciplinary Roles

Consulting: Operations, Tech-Driven Business Consulting

A dozen things

1. Optimise for Direction first, then Speed.
2. Focus is the new super-skill. To improve, manage your information diet.
3. Don't work on your weakness. Find your unique edge and leverage it to the fullest.
4. Identify and hone your skill stack.
5. Keep the fundamentals sharp.
6. Think in terms of 'body of work'.
7. Use the 'student I-card' to get entry and exposure to industry & institutions. Be driven by curiosity and absorb as much as you can.
8. Reading is not just another hobby. It's a fundamental skill. Start reading what you like, and keep it up till you like to read.
9. Learn the history and business of your field.
10. Look at knowledge as a lattice-work of concepts. Learn the interconnections and the hierarchies of concepts.
11. Learn to write with clarity.
12. Get mentors. Remember that some of your mentors may be long dead, but you can still learn from them!

Course LADDER

BY HIRANMAI MOHAN

FOURTH YEAR UNDERGRADUATE

To give a short introduction of myself, I'm currently going into my fourth year in MEMS, and I'm pursuing a major in MEMS with honors and a double minor in electrical engineering and chemistry. This, at first glance, might sound like a very odd combination, doing random courses in material science, chemistry and electrical engineering. Material science, being a highly diversified field, allows you to dive into a wide variety of research areas. My research interest is along the lines of semiconductor materials chemistry and processing.

Why take additional courses?

One unique thing that IITB offers over most other universities is allowing students to take courses from various departments ranging from 2xx to 8xx levels of difficulty. This enables one to pursue courses pertaining to their field of interest and develop stronger research fundamentals. It is true that mandatory coursework takes up a lot of your time and, in most cases, is exhausting enough to put you off from taking extra courses. But especially in the MEMS department, only the fundamentals are covered at the undergraduate level. Taking additional (usually senior-level) courses is beneficial to go towards a specific field.

Course Ladder

The initiative, termed "Course Ladder", is where you take a specific set of courses to design your own specialisation. There is no fixed way to do it. It can be done methodically if you have an idea of what you want to pursue or in a trial-and-error manner, at least in the initial stages when trying to explore different possibilities.

My Experience and Courses

My interest in pursuing research in materials science started while doing MM152, thanks to Prof. Parag Bhargava. Before joining the MEMS department as an undergraduate, I was already interested in chemistry. In fact, that is partly the reason that motivated me to join here. During pre-registration for the third semester, many of my seniors mentioned that pursuing an electrical minor would be beneficial since it is closely related to material science. I took up EE207 (Electronic Devices and Circuits) because it seemed the closest. I quite enjoyed the course and got into the field of semiconductors. I took up MM474 (Thin Film Science and Technology) to gain insight into the semiconductor fabrication process. I was more into experimental work than theoretical, so I wanted to take up courses on characterisation because basic knowledge of these techniques is a must for working in any lab. The courses I took up for the same are MM206 (Experimental Techniques in Material Science, 4th Sem) and CH521 (Interpretative Molecular Spectroscopy, 5th Sem).

I have also completed two other courses for my electrical minor, EE221 (Digital Electronics, 5th Sem) and EE229 (Signal Processing, 6th Sem), which are indirectly related to my interests. In my 6th Semester, I took up two more chemistry courses: CH504 (Computational Chemistry) and CH438 (Main Group Chemistry), to learn about other aspects of materials chemistry.

How did I benefit from this?

Apart from the theoretical learning aspect, I also realised that many of the concepts I learnt were pre-requisite background knowledge for the projects I was pursuing. Currently, I'm working on a project based on Solid State NMR for Energy Materials at University of Alberta, and it is directly related to parts of courses I have taken from all three departments. Initially, when I started taking these courses, it was more of a random choice (even now to some extent) but it soon began feeling interconnected.

General Pros and Con

Pros:

1. You get to explore new fields
2. It is not part of your CPI and most courses don't have a cutoff
3. You can drop the course if you don't find it interesting

Cons:

1. Requires extra time commitment, apart from core courses

Overall, it has been a great experience for me. I would personally recommend asking seniors about the courses and reading about the course content a bit before blindly taking it. Feel free to take as many courses as you're interested in. Happy learning!

Department Open Day

The Department of Metallurgical Engineering and Materials Science at IIT Bombay recently hosted an Open Day, welcoming over 400 school students from 10 different schools, spanning from 8th grade and above. This event, held on August 26, 2023, as part of the IOE outreach program, provided a unique opportunity for young minds to explore the fascinating world of metallurgical engineering and materials science.



In the lecture hall area of IIT Bombay, the Open Day was lively with students showing off demonstration, exhibits, and experiments. Led by enthusiastic department students, these activities highlighted the field's dynamism, stirring curiosity and fostering a love for learning



Among the standout demonstrations were explorations of concepts like scale and size, where students gained insights into the microscopic world and its implications on materials properties. Additionally, innovative experiments such as generating heat and electricity from motion, creating music from unconventional materials, and even designing their own superheroes through 3D printing captured the imaginations of participants.

As the day drew to a close, we extend our heartfelt gratitude to all faculty members, staff, volunteers, and students whose unwavering dedication and tireless efforts contributed to the resounding success of this event. Thanks to everyone's excitement and hard work, the Department Open Day was a big hit. It'll be remembered for a long time, and it's sure to keep inspiring and offering chances to people in the years ahead.



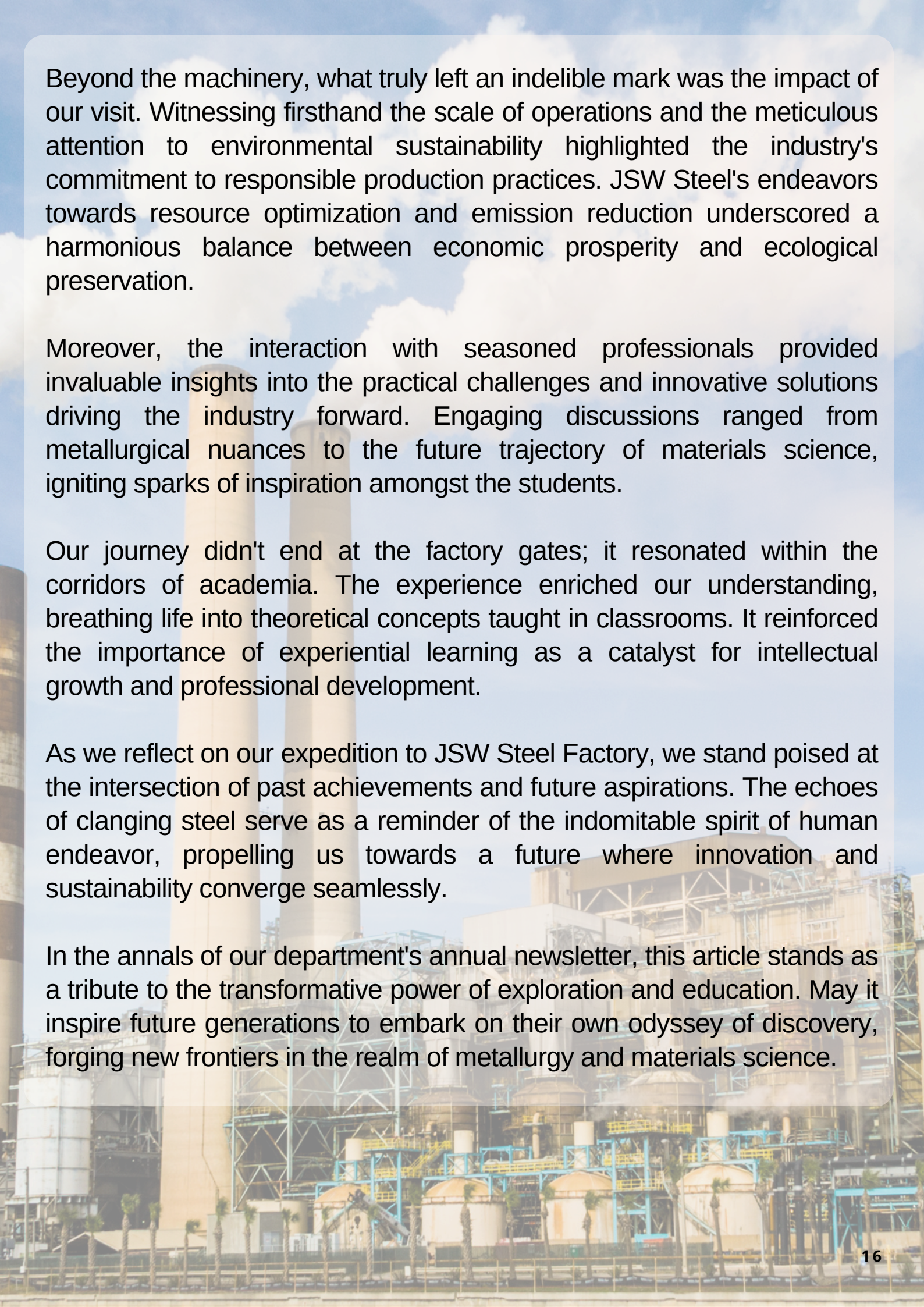
Industrial Visit

In a quest to bridge the gap between theoretical knowledge and practical applications, the students of Metallurgical Engineering and Materials Science embarked on a captivating journey to the JSW Steel Factory. Nestled amidst the industrial landscape, this behemoth of steel production stands as a testament to human ingenuity and technological advancement.

Accompanied by two esteemed professors, a group of thirty curious minds delved into the heart of steelmaking. The excursion was not merely a visit; it was an immersive experience, where every clang of machinery echoed the passion for metallurgy.

The focal point of our exploration was the blast furnace, a colossal structure where raw materials undergo a metamorphosis into molten iron. Standing in its presence, we were dwarfed by its sheer magnitude, yet captivated by its intricate workings. The heat, the roar, and the rhythmic flow of operations painted a vivid picture of industrial prowess.

As we traversed through the labyrinth of machinery, each component unveiled a unique facet of the steelmaking process. From the towering coke ovens to the precision of the rolling mills, every step was a symphony of precision engineering and scientific principles in action.



Beyond the machinery, what truly left an indelible mark was the impact of our visit. Witnessing firsthand the scale of operations and the meticulous attention to environmental sustainability highlighted the industry's commitment to responsible production practices. JSW Steel's endeavors towards resource optimization and emission reduction underscored a harmonious balance between economic prosperity and ecological preservation.

Moreover, the interaction with seasoned professionals provided invaluable insights into the practical challenges and innovative solutions driving the industry forward. Engaging discussions ranged from metallurgical nuances to the future trajectory of materials science, igniting sparks of inspiration amongst the students.

Our journey didn't end at the factory gates; it resonated within the corridors of academia. The experience enriched our understanding, breathing life into theoretical concepts taught in classrooms. It reinforced the importance of experiential learning as a catalyst for intellectual growth and professional development.

As we reflect on our expedition to JSW Steel Factory, we stand poised at the intersection of past achievements and future aspirations. The echoes of clanging steel serve as a reminder of the indomitable spirit of human endeavor, propelling us towards a future where innovation and sustainability converge seamlessly.

In the annals of our department's annual newsletter, this article stands as a tribute to the transformative power of exploration and education. May it inspire future generations to embark on their own odyssey of discovery, forging new frontiers in the realm of metallurgy and materials science.

Metals and Materials Summit

Metals and Materials Summit was a two day event organised on 24th-25th September 2022. The summit consisted of workshops, seminars panel discussions and competitions. The first day of the summit has workshop on Laser Applications and a Treasure Hunt over the academic area, apart from the event inauguration. The speaker of the workshop was Prof. Shobha Shukla who gave the participants a brief overview of Laser and its Applications.



The second day of the summit kicked off with Workshop Transport Electron Microscopy, by Prof. Salkalp SV Tatiparti. Workshop addressed the advantages and current limitations of TEM and demonstration along with explaining TEM and its working principle. Following the workshop was a Seminar Talk on the topic 'Materials Science using Neutrons' by Dr. S M Yusuf, HOD Physics BARC. Seminars had a big participation from the students and professors. Another event which was carried out in the first half of the day was the Panel Discussion on Emerging Semiconductors Technologies. The panel consisted of Gopi Chandran (Deputy Director R&D Applied Materials), Prof. Rajiv Dusane (MEMS, IIT Bombay), Prof. Abhijeet Sangle (MEMS, IIT Bombay), Prof. Triratna Muneshwar (MEMS, IIT Bombay). This discussion was mainly focused on the topics of role of semiconductors in our day-to-day life, Semiconductor technologies- photovoltaics, and logic/memory, present status of the field & its future aspects and the roles that India is setting for in this field.

The second half of the day was started with a Poster Presentation competition, in which PhD students presented their research with the help of posters. This was followed with a panel discussion on Sustainability of Materials, paneled by Rishi Pathania (Head Sustainability - UPL), Nirbhay Lumde (Head CSR APAC - CGI), and Prof. Pradip Kalbar(CUSE, IIT Bombay). The discussion was done on Sustainability of metals and their roles in achieving net zero & circular economy, policies adopted by companies and government to enhance material performance and make it more sustainable and recent developments in sustainable chemicals and materials.

This was followed by the 'Met Wiz' competition, a quiz competition on Metallurgy knowledge next panel discussion in line was on emerging Market of Induction Furnace in Indian Steel Industry by Nitin Kabra (Director – Bhagyalakshmi), Vishva Bandhu (Sr. Deputy Director - NISST), Prof. Asim Tewari (Mechanical Engineering, IIT Bombay), S V Panse (Chief of Commercial, Stainless Steel, MUKAND).

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The final panel discussion of the Materials Summit was on 'Future of Indian Steel Industry'. The discussion had panelists Gajraj Singh Rathore (COO - JSW), Naveen Ahlawat (Head Green Initiatives - JSPL) and Prof. N N Vishwanathan (HoD, MEMS, IIT Bombay). The panel discussed about carbon credits, Green technologies, Policies needed for achieving our net zero Goals, recent developments in legacy companies, Circular Economy, Energy requirements and raw materials availability.

This concluded the first ever MMA Summit by MEMS, IIT Bombay. The summit focused on bridging gap between Information gap between Research- driven academia, industries, policymakers, and machinery manufacturers, by bringing them together.

Materials Summits saw participation of 700+ engineering professionals, 20+ guest speakers & 50+ industry veterans. The meaningful discussions with Mr. Gajraj S. Rathore (COO - JSW) and Naveen Ahlawat (Head Green Initiatives - JSPL) about the role of hydrogen in carbon-neutral steel while discussion its limitations. Rathore Sir's presentation on summarizing our path to green steel by 2070 was detailed & data-driven, with an extensive techno-economical analysis of the most promising technologies.



END NOTE

Abhinandan Kumar Nikhil Department General Secretary 2023-24



Dear Readers,

It brings me great joy to present to you the latest edition of our department newsletter, Dhatuki. Our department has a rich history, dating back to the very establishment of our esteemed institution in 1958. Over the years, we have continually strived for excellence in all aspects, be it academics, research, administration, or laboratory work.

As the General Secretary of the department, I am immensely proud of the strides we have made in fostering a culture of excellence. It is with this spirit that I extend a warm message to each and every student within our department:

"As students of this department, we're privileged to belong to a longstanding fraternity that offers access to a wealth of resources, including the guidance of esteemed professors, cutting-edge laboratories, experienced seniors, and top-notch facilities. It is incumbent upon us to capitalize on these resources and actively engage in learning from them. The department's commitment to enhancing the student learning experience is evident through recent curriculum changes. I encourage you to explore beyond the confines of textbooks and immerse yourselves in the realms of research and entrepreneurship. In this edition of Dhatuki, we aim to provide you with a diverse array of insights, including interviews with our professors on their latest research endeavors and updates on the latest advancements in the field of Material Sciences."

I extend my heartfelt gratitude to all the professors, seniors, and alumni who have graciously shared their expertise and insights with us. Special thanks are also due to our esteemed HoD, Prof. N. Vishwanathan, for his invaluable guidance and unwavering support, without which this publication would not have been possible.

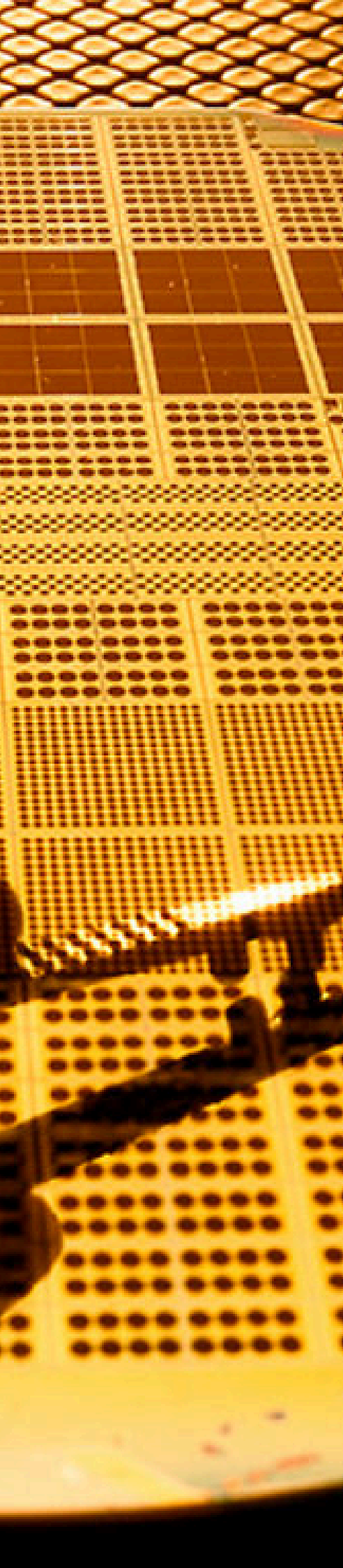
Furthermore, I commend the tireless efforts of the team, led by Bhumika Bhangale. Their dedication and hard work have culminated in the successful publication of this edition, aimed at benefitting both current and future students of the department.

I trust that you will find this edition of Dhatuki both insightful and enriching.

Thank you for your continued support!

Warm regards,

Abhinandan Kumar Nikhil



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